1 Mathematical and Economic Foundations

A Mathematical Preliminaries

1 Unconstrained Optimization
2 Constrained Optimization

B Consumer Optimization

1 Graphical Analysis
2 Algebraic Analysis
3 The Time Dimension

Mathematical Preliminaries

Unconstrained Optimization

$$\max_x F(x)$$

Constrained Optimization

$$\max_x F(x) \text{ subject to } c \geq G(x)$$

Unconstrained Optimization

To find the value of x that solves

$$\max_x F(x)$$

you can:

1. Try out every possible value of x.
2. Use calculus.

Since search could take forever, let's use calculus instead.

Unconstrained Optimization

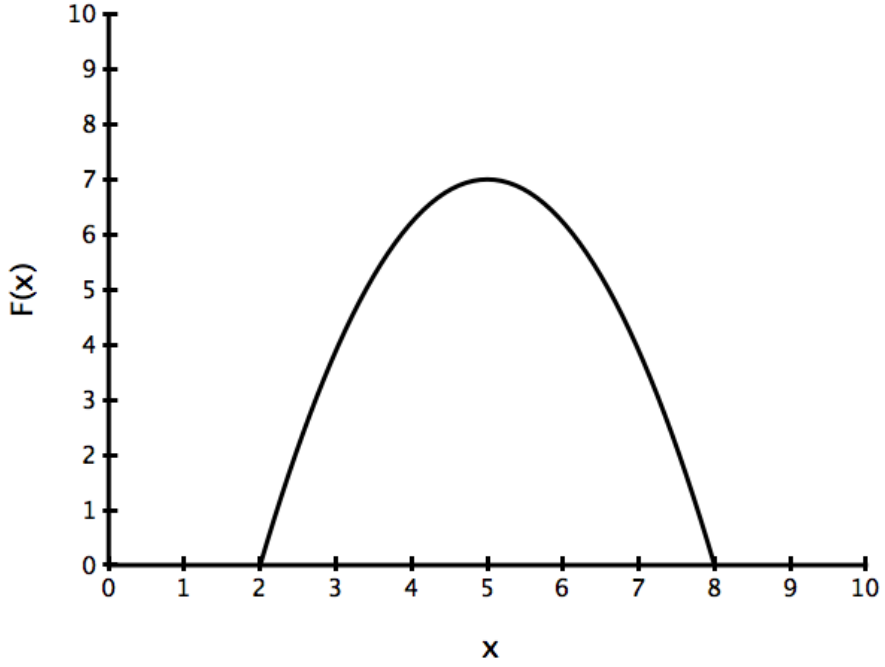
Theorem If x^* solves

$$\max_x F(x),$$

then x^* is a critical point of F, that is,

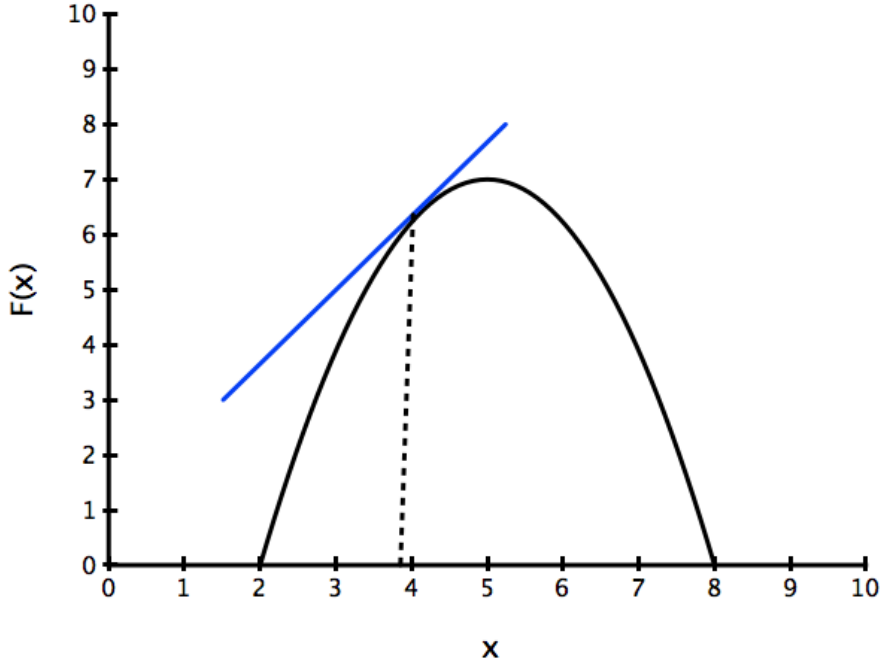
$$F'(x^*) = 0.$$

Unconstrained Optimization



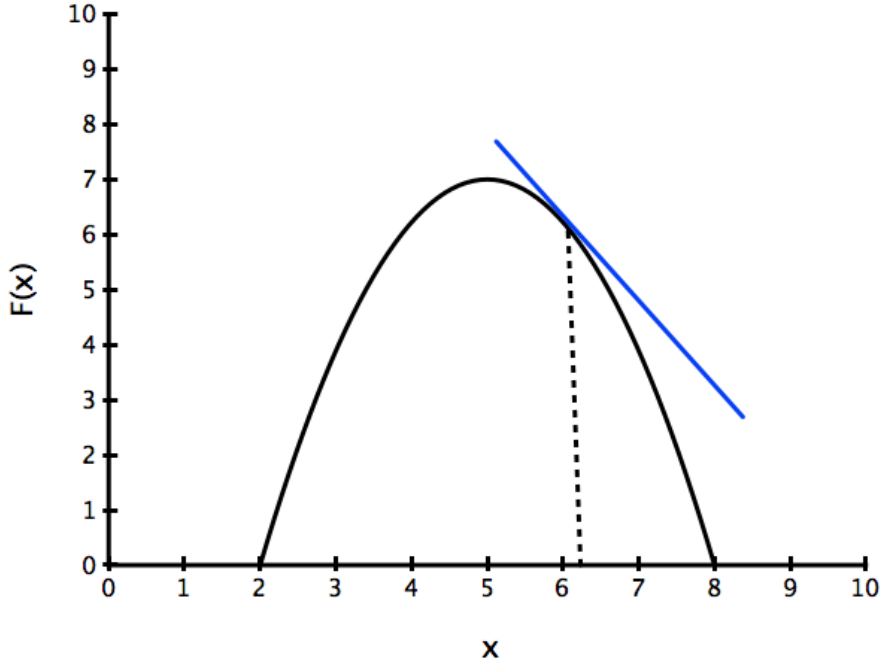
$F(x)$ maximized at $x^* = 5$

Unconstrained Optimization



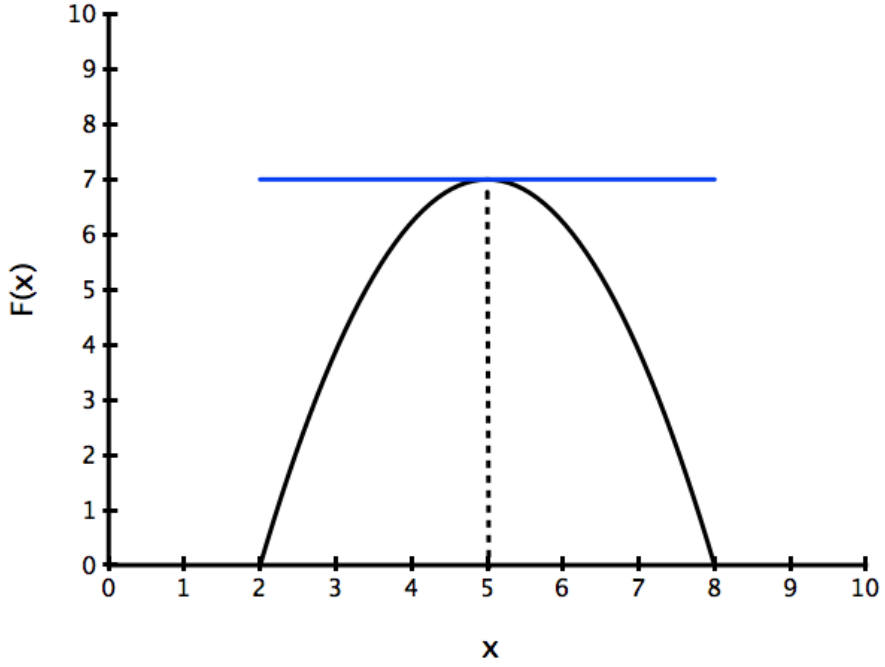
$F'(x) > 0$ when $x < 5$. $F(x)$ can be increased by increasing x.

Unconstrained Optimization



$F'(x) < 0$ when $x > 5$. $F(x)$ can be increased by decreasing x.

Unconstrained Optimization



$F'(x) = 0$ when $x = 5$. $F(x)$ is maximized.

Unconstrained Optimization

Theorem If x^* solves

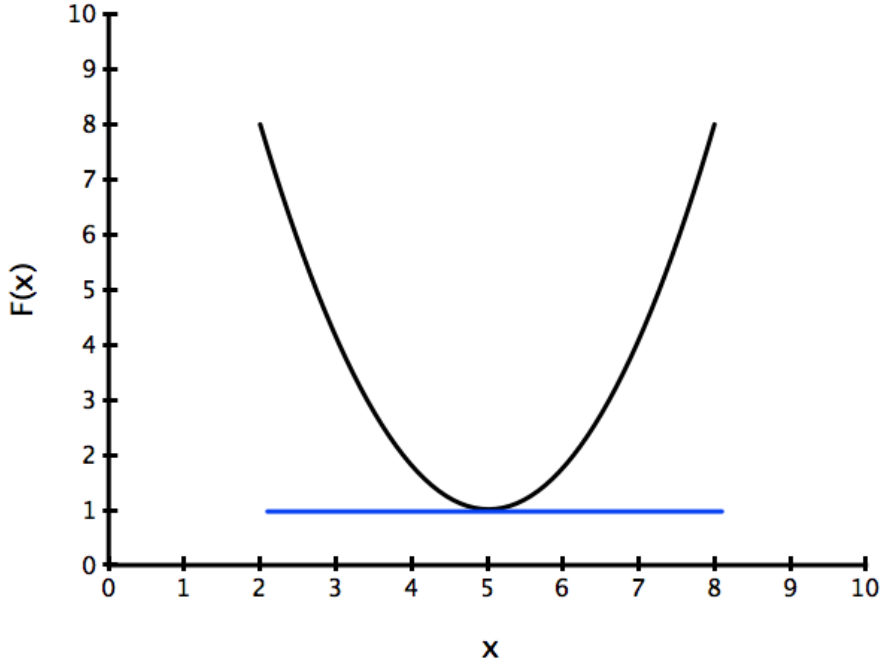
$$\max_x F(x),$$

then x^* is a critical point of F, that is,

$$F'(x^*) = 0.$$

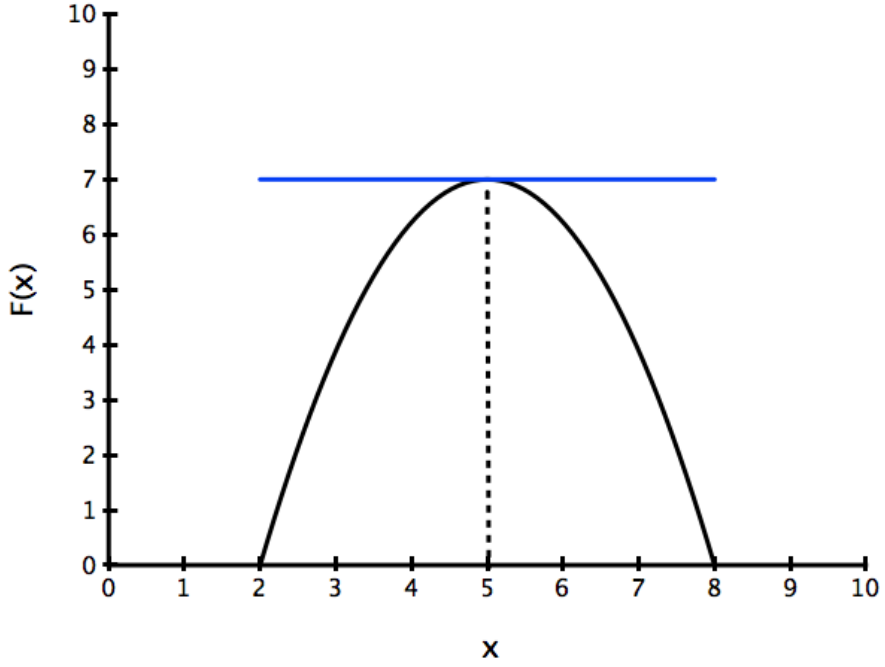
Note that the same first-order necessary condition $F'(x^*) = 0$ also characterizes a value of x^* that minimizes $F(x)$.

Unconstrained Optimization



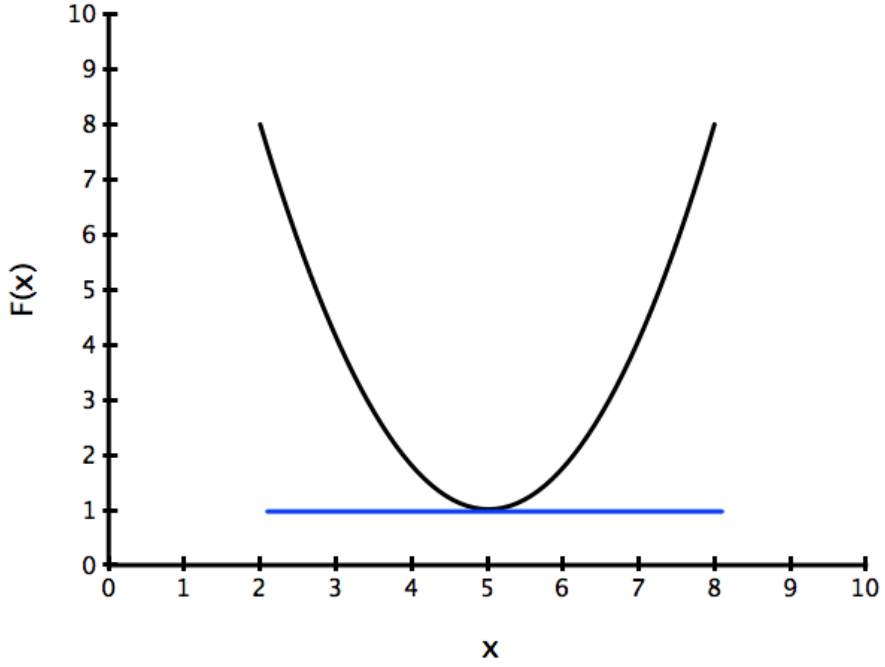
$F'(x) = 0$ when $x = 5$. $F(x)$ is minimized.

Unconstrained Optimization



$F'(x) = 0$ and $F''(x) < 0$ when $x = 5$. $F(x)$ is maximized.

Unconstrained Optimization



$F'(x) = 0$ and $F''(x) > 0$ when $x = 5$. $F(x)$ is minimized.

Unconstrained Optimization

Theorem If x^* solves

$$\max_x F(x),$$

then x^* is a critical point of F, that is,

$$F'(x^*) = 0.$$

Unconstrained Optimization

Theorem If

$$F'(x^*) = 0 \text{ and } F''(x^*) < 0,$$

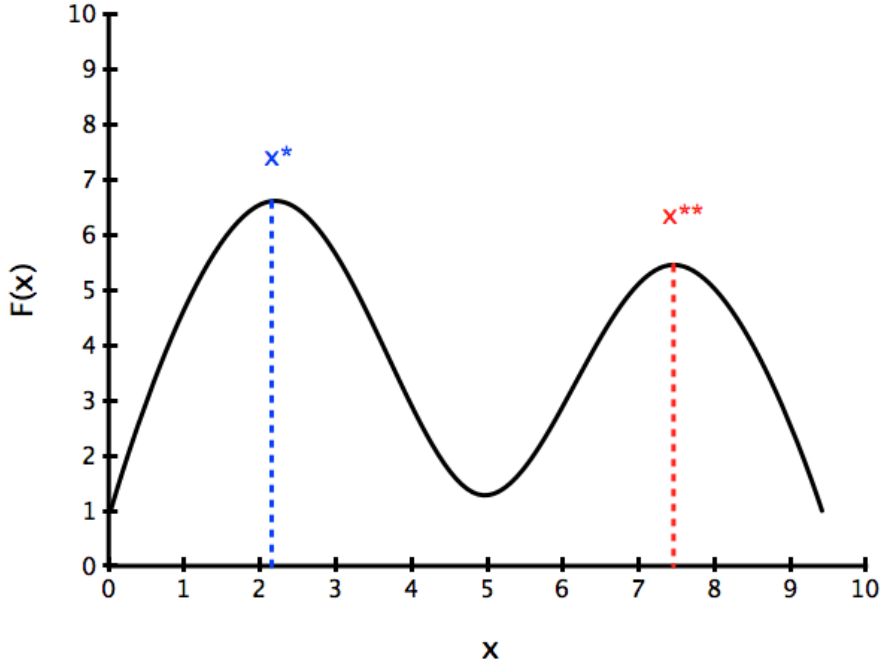
then x^* solves

$$\max_x F(x)$$

(at least locally).

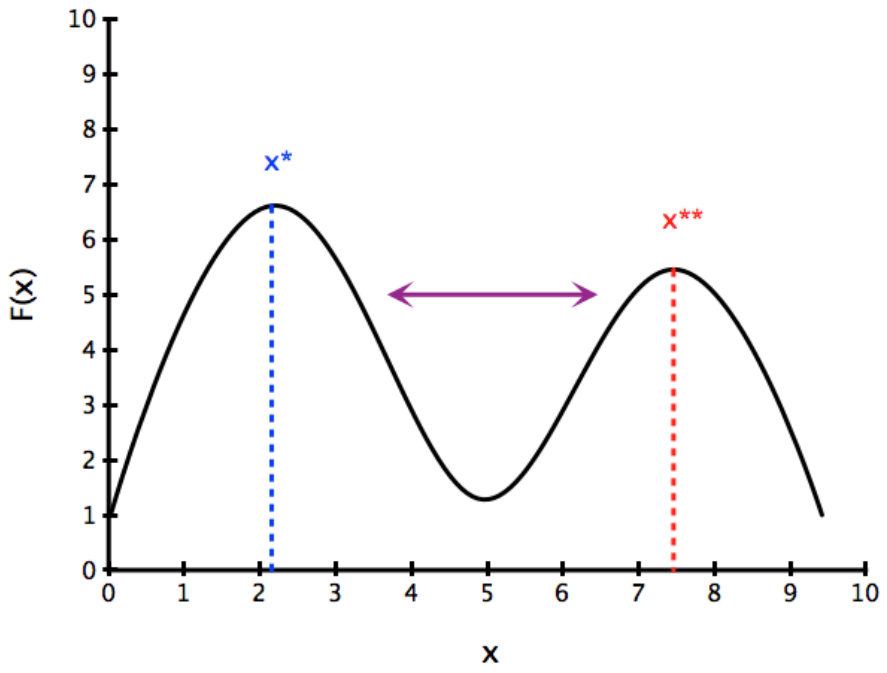
The first-order condition $F'(x^*) = 0$ and the second-order condition $F''(x^*) < 0$ are sufficient conditions for the value of x that (locally) maximizes $F(x)$.

Unconstrained Optimization



$F'(x^{**}) = 0$ and $F''(x^{**}) < 0$ at the local maximizer x^{**} and $F'(x^{*}) = 0$ and $F''(x^{*}) < 0$ at the global maximizer x^{*}.

Unconstrained Optimization



$F'(x^{**}) = 0$ and $F''(x^{**}) < 0$ at the local maximizer x^{**} and $F'(x^*) = 0$ and $F''(x^*) < 0$ at the global maximizer x^*, but $F''(x) > 0$ in between x^* and x^{**}.

Unconstrained Optimization

Theorem If

$$F'(x^*) = 0$$

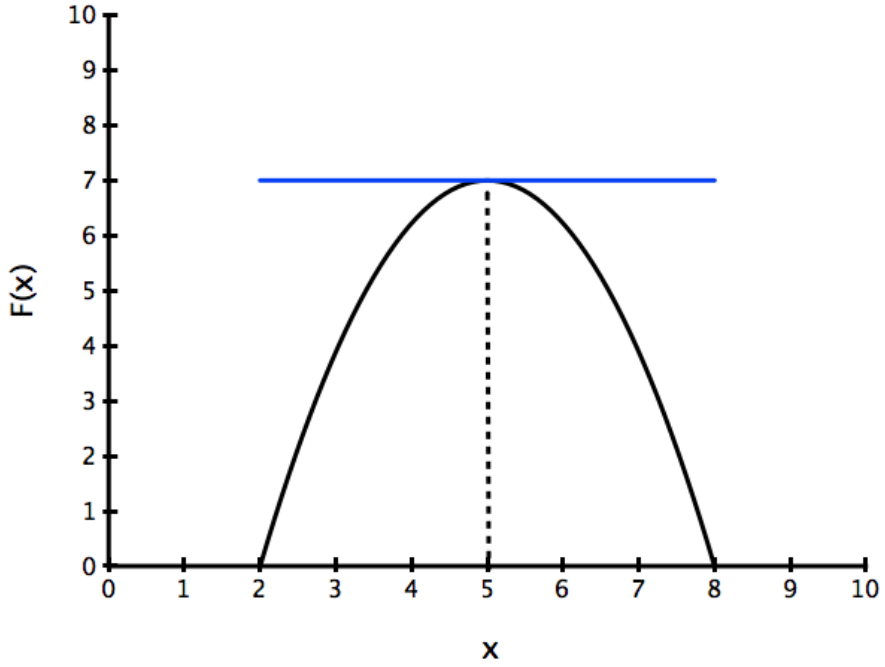
and

$$F''(x) < 0 \text{ for all } x \in \mathbb{R},$$

then x^* solves

$$\max_x F(x).$$

Unconstrained Optimization



$F''(x) < 0$ for all $x \in \mathbb{R}$ and $F'(5) = 0$. $F(x)$ is maximized when $x = 5$.

Unconstrained Optimization

If $F''(x) < 0$ for all $x \in \mathbb{R}$, then the function F is concave.

When F is concave, the first-order condition $F'(x^*) = 0$ is both necessary and sufficient for the value of x that maximizes $F(x)$.

And, as we are about to see, concave functions arise frequently and naturally in economics and finance.

Unconstrained Optimization: Example 1

Consider the problem

$$\max_x \left(-\frac{1}{2}\right)(x-\tau)^2,$$

where τ is a number ($\tau \in \mathbb{R}$) that we might call the "target."

The first-order condition

$$-(x^* - \tau) = 0$$

leads us immediately to the solution: $x^* = \tau$.

Unconstrained Optimization: Example 2

Consider maximizing a function of three variables:

$$\max_{x_1, x_2, x_3} F(x_1, x_2, x_3)$$

Even if each variable can take on only 1,000 values, there are one billion possible combinations of (x_1, x_2, x_3) to search over!

This is an example of what Richard Bellman (US, 1920-1984) called the "curse of dimensionality."

Unconstrained Optimization: Example 2

Consider the problem:

$$\max_{x_1,x_2,x_3} \left(-\frac{1}{2}\right)(x_1-\tau)^2 + \left(-\frac{1}{2}\right)(x_2-x_1)^2 + \left(-\frac{1}{2}\right)(x_3-x_2)^2.$$

Now the three first-order conditions

$$-(x_1^* - \tau) + (x_2^* - x_1^*) = 0$$
$$-(x_2^* - x_1^*) + (x_3^* - x_2^*) = 0$$
$$-(x_3^* - x_2^*) = 0$$

lead us to the solution: $x_1^* = x_2^* = x_3^* = \tau$.

Constrained Optimization

To find the value of x that solves

$$\max_x F(x) \text{ subject to } c \geq G(x)$$

you can:

1. Try out every possible value of x.
2. Use calculus.

Since search could take forever, let's use calculus instead.

Constrained Optimization

A method for solving constrained optimization problems like

$$\max_x F(x) \text{ subject to } c \geq G(x)$$

was developed by Joseph-Louis Lagrange (France/Italy, 1736-1813) and extended by Harold Kuhn (US, b.1925) and Albert Tucker (US, 1905-1995).

Constrained Optimization

Associated with the problem:

$$\max_x F(x) \text{ subject to } c \geq G(x)$$

Define the Lagrangian

$$L(x, \lambda) = F(x) + \lambda[c - G(x)],$$

where λ is the Lagrange multiplier.

Constrained Optimization

$$L(x, \lambda) = F(x) + \lambda[c - G(x)],$$

Theorem (Kuhn-Tucker) If x^* maximizes $F(x)$ subject to $c \geq G(x)$, then there exists a value $\lambda^* \geq 0$ such that, together, x^* and λ^* satisfy the first-order condition

$$F'(x^*) - \lambda^* G'(x^*) = 0$$

and the complementary slackness condition

$$\lambda^*[c - G(x^*)] = 0.$$

Constrained Optimization

In the case where $c > G(x^*)$, the constraint is non-binding. The complementary slackness condition

$$\lambda^*[c - G(x^*)] = 0$$

requires that $\lambda^* = 0$.

And the first-order condition

$$F'(x^*) - \lambda^* G'(x^*) = 0$$

requires that $F'(x^*) = 0$.

Constrained Optimization

In the case where $c = G(x^*)$, the constraint is binding. The complementary slackness condition

$$\lambda^*[c - G(x^*)] = 0$$

puts no further restriction on $\lambda^* \geq 0$.

Now the first-order condition

$$F'(x^*) - \lambda^* G'(x^*) = 0$$

requires that $F'(x^*) = \lambda^* G'(x^*)$.

Constrained Optimization: Example 1

For the problem

$$\max_x \left(-\frac{1}{2}\right)(x-5)^2 \text{ subject to } 7 \geq x,$$

$F(x) = (-1/2)(x-5)^2$, $c = 7$, and $G(x) = x$. The Lagrangian is

$$L(x, \lambda) = \left(-\frac{1}{2}\right)(x-5)^2 + \lambda(7-x).$$

Constrained Optimization: Example 1

With

$$L(x, \lambda) = \left(-\frac{1}{2}\right)(x-5)^2 + \lambda(7-x),$$

the first-order condition

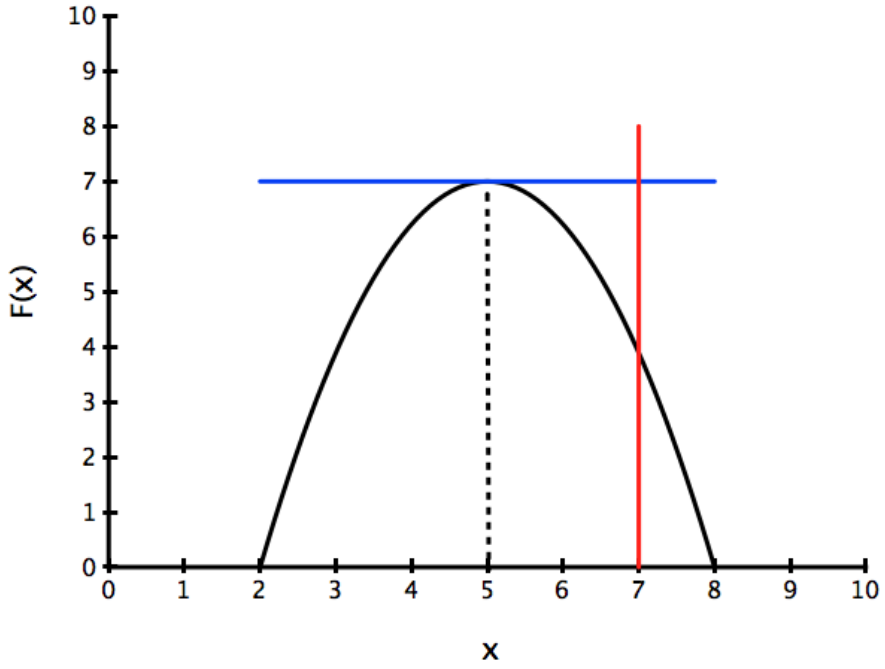
$$-(x^* - 5) - \lambda^* = 0$$

and the complementary slackness condition

$$\lambda^*(7 - x^*) = 0$$

are satisfied with $x^* = 5$, $F'(x^*) = 0$, $\lambda^* = 0$, and $7 > x^*$.

Constrained Optimization: Example 1



Here, the solution has $F'(x^*) = 0$ since the constraint is nonbinding.

Constrained Optimization: Example 2

For the problem

$$\max_x \left(-\frac{1}{2}\right)(x-5)^2 \text{ subject to } 4 \geq x,$$

$F(x) = (-1/2)(x-5)^2$, $c = 4$, and $G(x) = x$. The Lagrangian is

$$L(x, \lambda) = \left(-\frac{1}{2}\right)(x-5)^2 + \lambda(4-x).$$

Constrained Optimization: Example 2

With

$$L(x, \lambda) = \left(-\frac{1}{2}\right)(x-5)^2 + \lambda(4-x),$$

the first-order condition

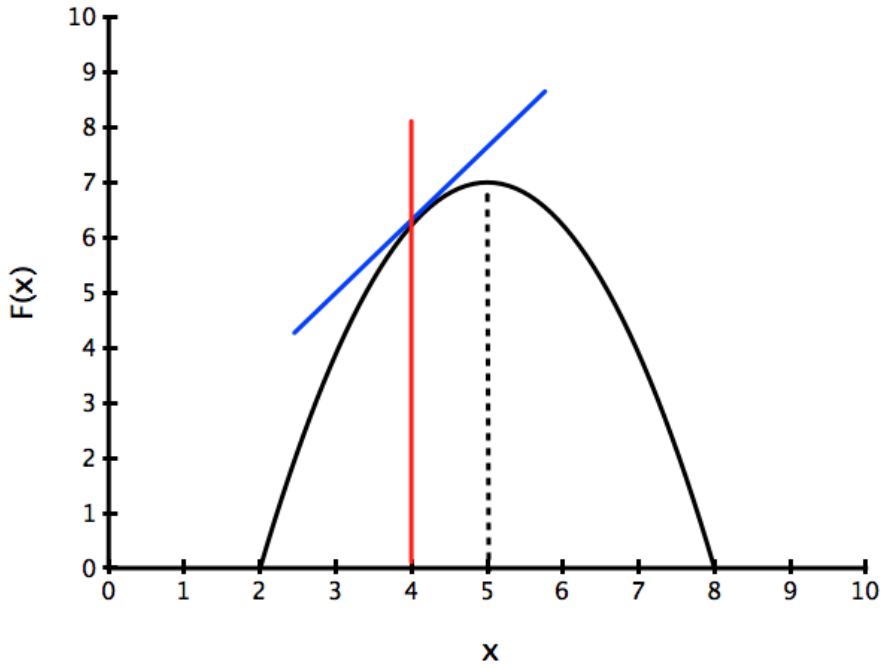
$$-(x^* - 5) - \lambda^* = 0$$

and the complementary slackness condition

$$\lambda^*(4 - x^*) = 0$$

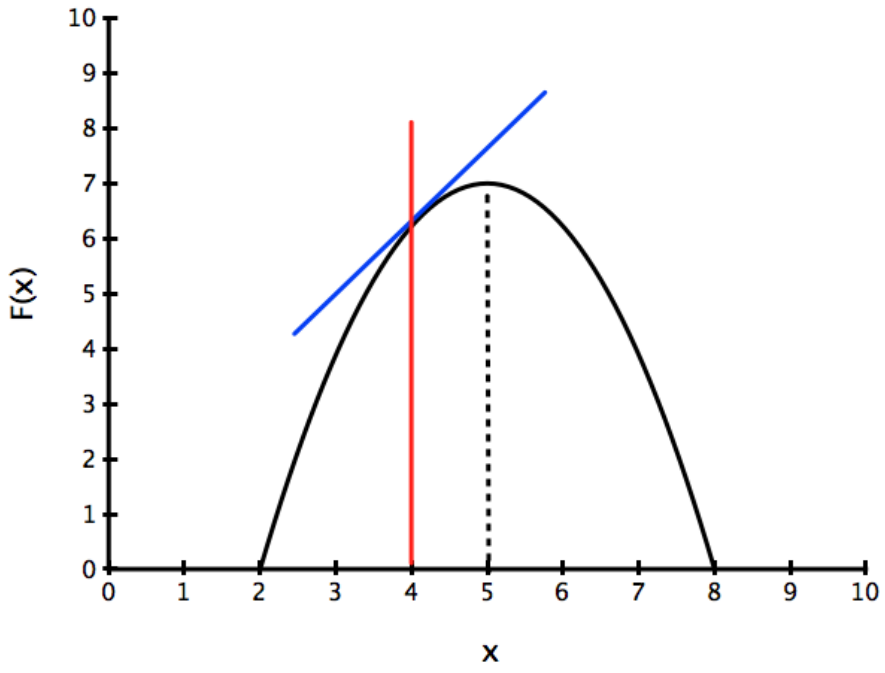
are satisfied with $x^* = 4$ and $F'(x^*) = \lambda^* = 1 > 0$.

Constrained Optimization: Example 2



Here, the solution has $F'(x^*) = \lambda^* G'(x^*) > 0$ since the constraint is binding. $F'(x^*) > 0$ indicates that we'd like to increase the value of x, but the constraint won't let us.

Constrained Optimization: Example 2



With a binding constraint, $F'(x^*) \neq 0$ but $F'(x^*) - \lambda^* G'(x^*) = 0$. The value x^* that solves the problem is a critical point, not of the objective function $F(x)$, but instead of the entire Lagrangian $F(x) + \lambda[c - G(x)]$.

Consumer Optimization

1. Graphical Analysis
2. Algebraic Analysis
3. Time Dimension
4. Risk Dimension

Consumer Optimization

Alfred Marshall, *Principles of Economics*, 1890. – supply and demand

Francis Edgeworth, *Mathematical Psychics*, 1881.

Vilfredo Pareto, *Manual of Political Economy*, 1906. – indifference curves

Consumer Optimization

John Hicks, *Value and Capital*, 1939. – wealth and substitution effects

Paul Samuelson, *Foundations of Economic Analysis*, 1947. – mathematical reformulation

Irving Fisher, *The Theory of Interest*, 1930. – intertemporal extension.

Consumer Optimization

Gerard Debreu, *Theory of Value*, 1959.

Kenneth Arrow, "The Role of Securities in the Optimal Allocation of Risk Bearing," *Review of Economic Studies*, 1964.

Extensions to include risk and uncertainty.

Consumer Optimization: Graphical Analysis

Consider a consumer who likes two goods: apples and bananas.

$$Y = \text{income}$$
$$c_a = \text{consumption of apples}$$
$$c_b = \text{consumption of bananas}$$
$$p_a = \text{price of an apple}$$
$$p_b = \text{price of a banana}$$

The consumer's budget constraint is

$$Y \geq p_a c_a + p_b c_b$$

Consumer Optimization: Graphical Analysis

So long as the consumer always prefers more to less, the budget constraint will always bind:

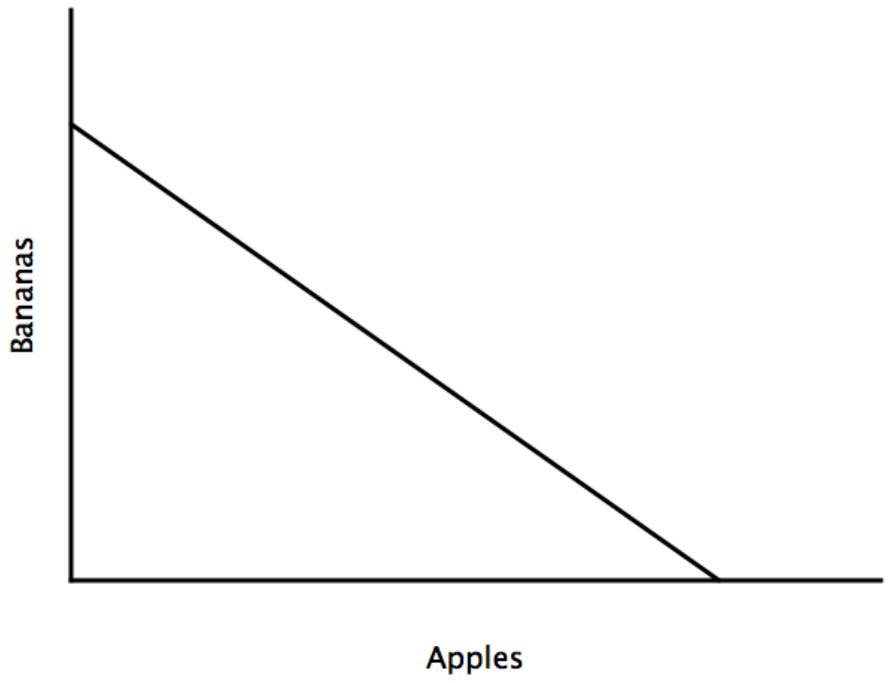
$$Y = p_a c_a + p_b c_b$$

or

$$c_b = \frac{Y}{p_b} - \left(\frac{p_a}{p_b}\right) c_a$$

Which shows that the graph of the budget constraint will be a straight line with slope $-(p_a/p_b)$ and intercept Y/p_b.

Consumer Optimization: Graphical Analysis



The budget constraint is a straight line with slope $-(p_a/p_b)$ and intercept Y/p_b.

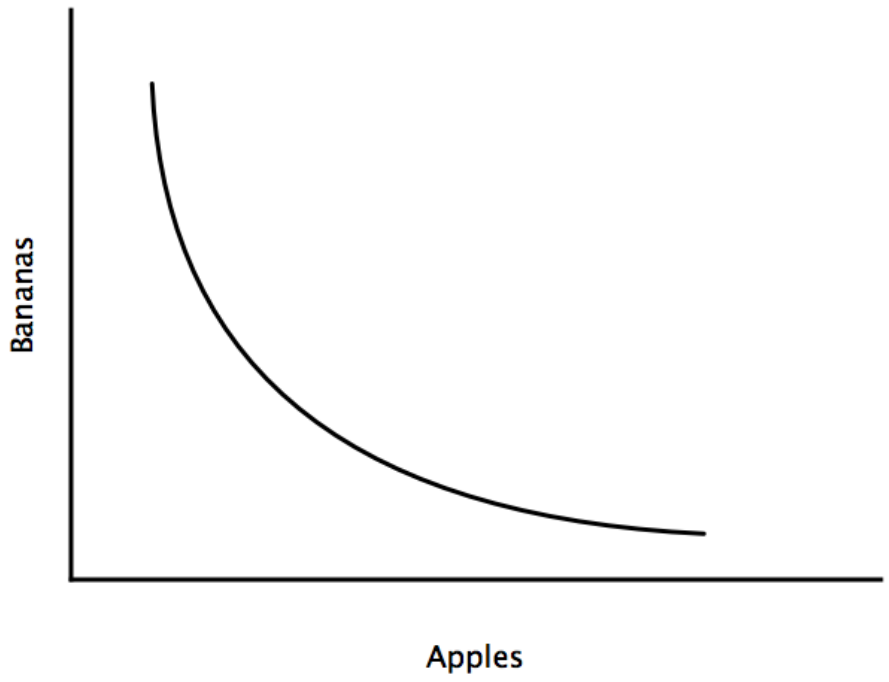
Consumer Optimization: Graphical Analysis

The budget constraint describes the consumer's market opportunities.

Francis Edgeworth (Ireland, 1845-1926) and Vilfredo Pareto (Italy, 1848-1923) were the first to use indifference curves to describe the consumer's preferences.

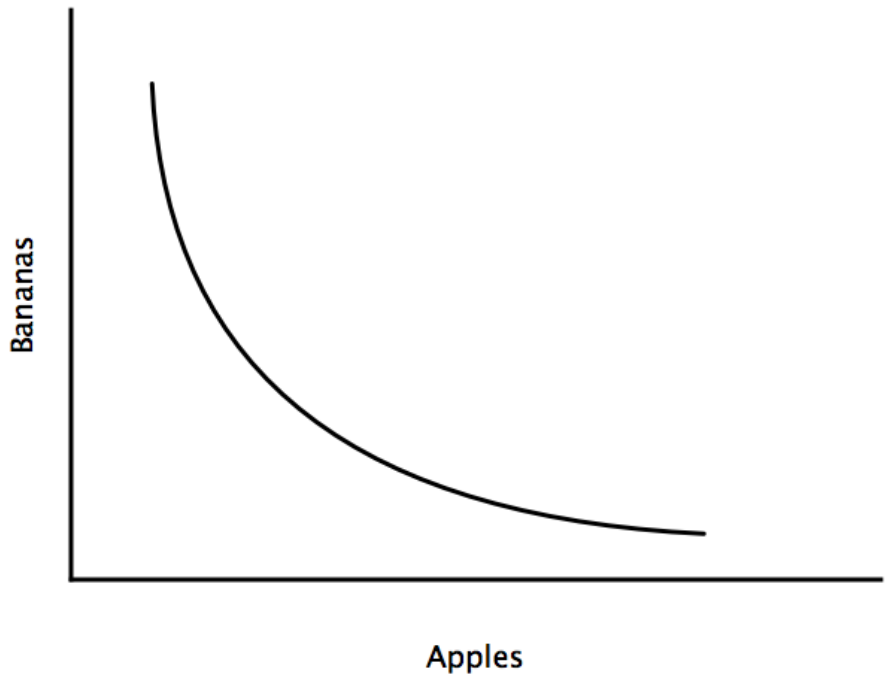
Each indifference curve traces out a set of combinations of apples and bananas that give the consumer a given level of utility or satisfaction.

Consumer Optimization: Graphical Analysis



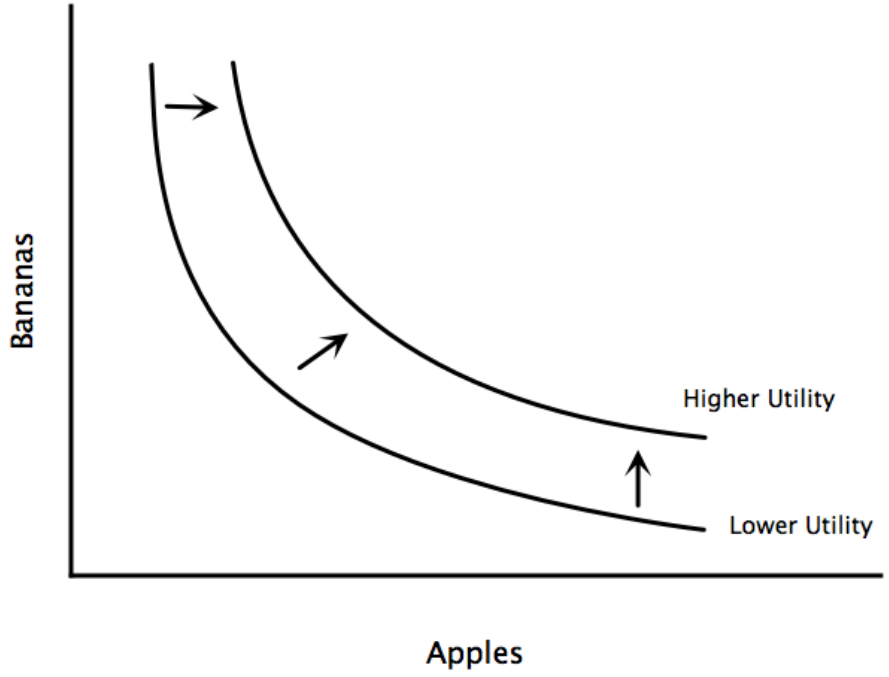
Each indifference curve traces out a set of combinations of apples and bananas that give the consumer a given level of utility.

Consumer Optimization: Graphical Analysis



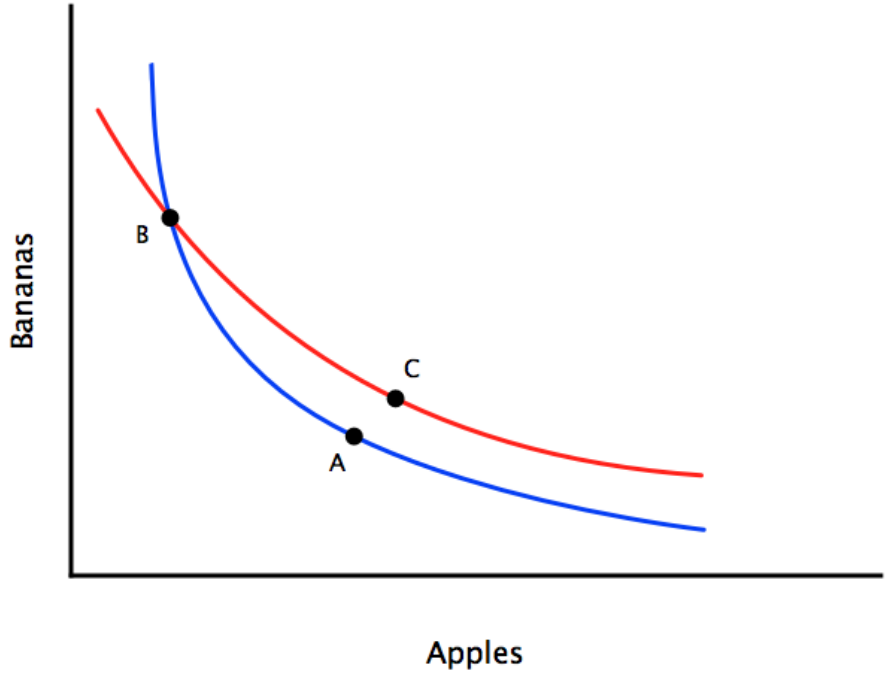
Each indifference curve slopes down, since the consumer requires more apples to compensate for a loss of bananas and more bananas to compensate for a loss of apples, if more is preferred to less.

Consumer Optimization: Graphical Analysis



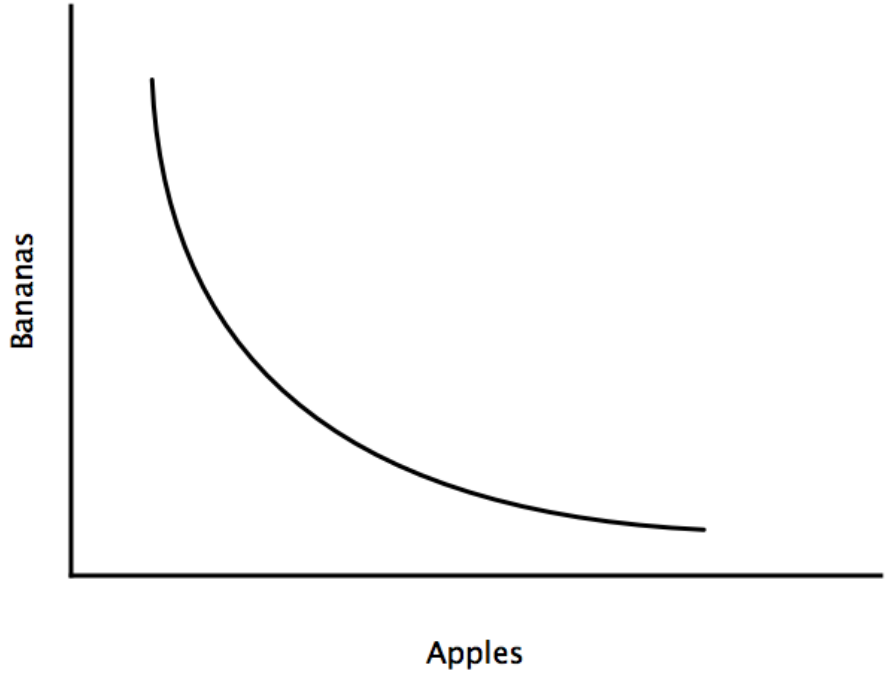
Indifference curves farther away from the origin represent higher levels of utility, if more is preferred to less.

Consumer Optimization: Graphical Analysis



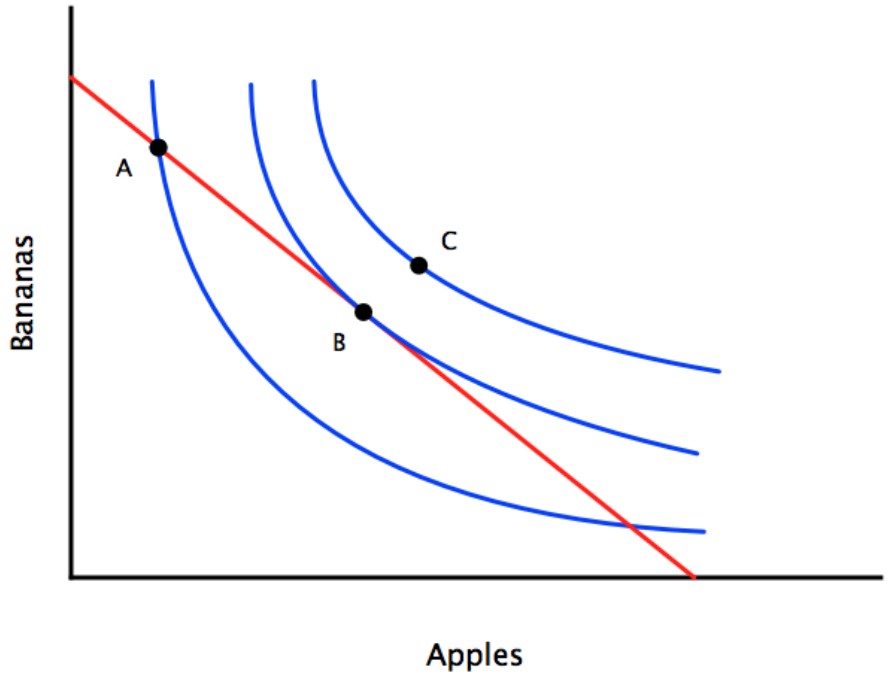
A and B yield the same level of utility, and B and C yield the same level of utility, but C is preferred to A if more is preferred to less. Indifference curves cannot intersect.

Consumer Optimization: Graphical Analysis



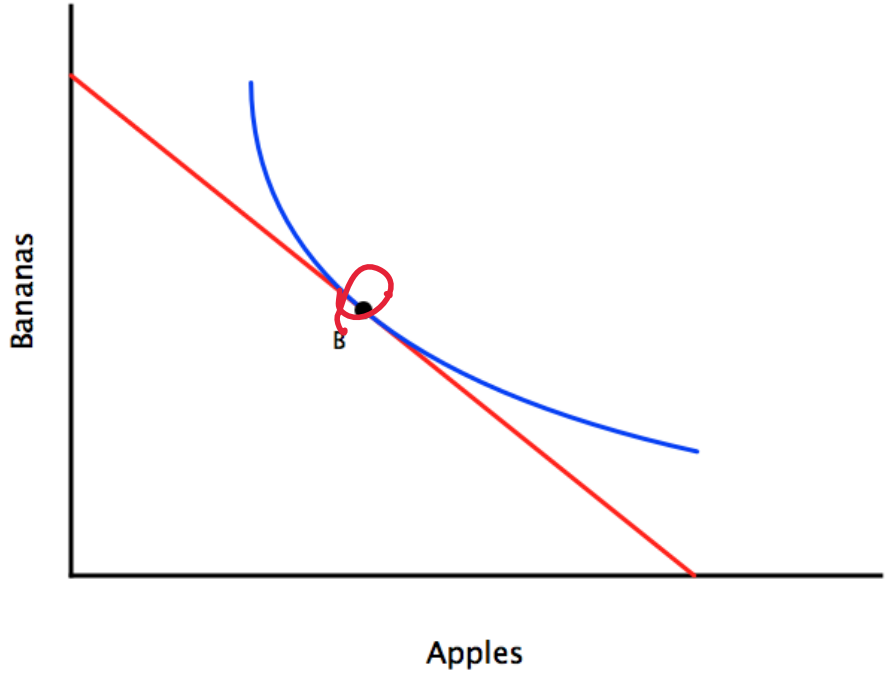
Indifference curves are convex to the origin if consumers have a preference for diversity.

Consumer Optimization: Graphical Analysis



A is suboptimal and C is infeasible. B is optimal.

Consumer Optimization: Graphical Analysis



At B, the optimal choice, the indifference curve is tangent to the budget constraint.

Consumer Optimization: Graphical Analysis

Recall that the budget constraint

$$Y = p_a c_a + p_b c_b$$

or

$$c_b = \frac{Y}{p_b} - \left(\frac{p_a}{p_b}\right) c_a$$

has slope $-(p_a/p_b)$.

Consumer Optimization: Graphical Analysis

Suppose that the consumer's preferences are also described by the utility function

$$u(c_a) + \beta u(c_b).$$

The function u is increasing, with $u'(c) > 0$, so that more is preferred to less, and concave, with $u''(c) < 0$, so that marginal utility falls as consumption rises.

The parameter β measures how much more (if $\beta > 1$) or less (if $\beta < 1$) the consumer likes bananas compared to apples.

Consumer Optimization: Graphical Analysis

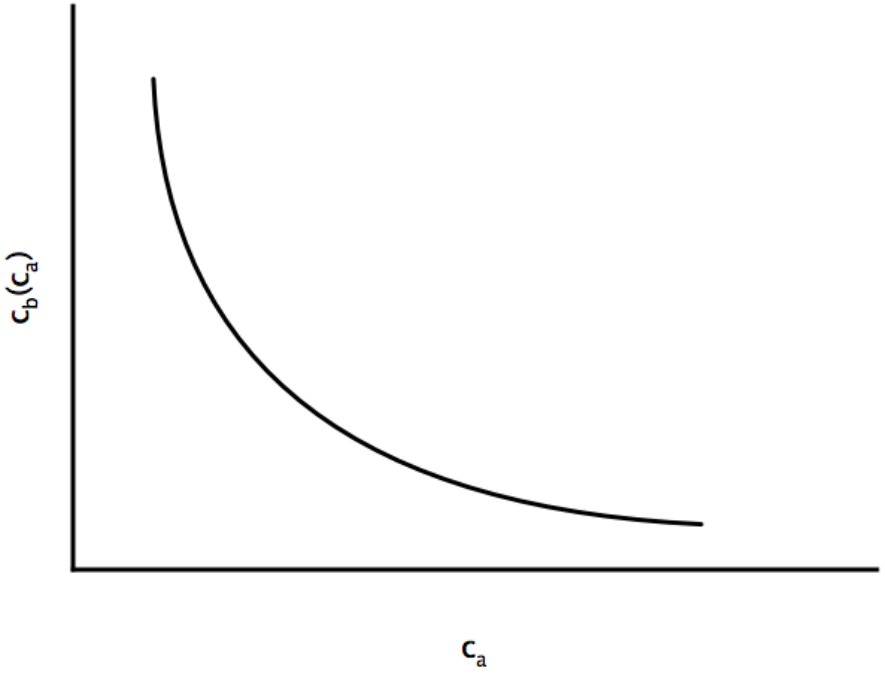
Since an indifference curve traces out the set of (c_a, c_b) combinations that yield a given level of utility $\bar{U}$, the equation for an indifference curve is

$$\bar{U} = u(c_a) + \beta u(c_b).$$

Use this equation to define a new function, $c_b(c_a)$, describing the number of bananas needed, for each number of apples, to keep the consumer on this indifference curve:

$$\bar{U} = u(c_a) + \beta u[c_b(c_a)].$$

Consumer Optimization: Graphical Analysis



The function $c_b(c_a)$ satisfies $\bar{U} = u(c_a) + \beta u[c_b(c_a)]$.

Consumer Optimization: Graphical Analysis

Differentiate both sides of

$$\bar{U} = u(c_a) + \beta u[c_b(c_a)]$$

to obtain

$$0 = u'(c_a) + \beta u'[c_b(c_a)]c_b'(c_a)$$

or

$$c_b'(c_a) = -\frac{u'(c_a)}{\beta u'[c_b(c_a)]}.$$

Consumer Optimization: Graphical Analysis

This last equation,

$$c_b'(c_a) = -\frac{u'(c_a)}{\beta u'[c_b(c_a)]},$$

written more simply as

$$c_b'(c_a) = -\frac{u'(c_a)}{\beta u'(c_b)},$$

measures the slope of the indifference curve: the consumer's marginal rate of substitution.

Consumer Optimization: Graphical Analysis

Thus, the tangency of the budget constraint and indifference curve can be expressed mathematically as

$$\frac{p_a}{p_b} = \frac{u'(c_a)}{\beta u'(c_b)}.$$

The marginal rate of substitution equals the relative prices.

Consumer Optimization: Graphical Analysis

Returning to the more general expression

$$c_b'(c_a) = -\frac{u'(c_a)}{\beta u'[c_b(c_a)]},$$

we can see that $c_b'(c_a) < 0$, so that the indifference curve is downward-sloping, so long as the utility function u is strictly increasing, that is, if more is preferred to less.

Consumer Optimization: Graphical Analysis

$$c_b'(c_a) = -\frac{u'(c_a)}{\beta u'[c_b(c_a)]}$$

Differentiating again yields

$$c_b''(c_a) = -\frac{\beta u'[c_b(c_a)]u''(c_a) - u'(c_a)\beta u''[c_b(c_a)]c_b'(c_a)}{\{\beta u'[c_b(c_a)]\}^2},$$

which is positive if u is strictly increasing (more is preferred to less) and concave (diminishing marginal utility). In this case, the indifference curve will be convex. Again, we see how concave functions have mathematical properties and economic implications that we like.

Consumer Optimization: Algebraic Analysis

Graphical analysis works fine with two goods.

But what about three goods? That depends on how good an artist you are!

And what about four or more goods? Our universe won't accommodate a graph like that!

But once again, calculus makes it easier!

Consumer Optimization: Algebraic Analysis

Consider a consumer who likes three goods:

$$Y = \text{income}$$
$$c_i = \text{consumption of goods } i = 0, 1, 2$$
$$p_i = \text{price of goods } i = 0, 1, 2$$

Suppose the consumer's utility function is

$$u(c_0) + \alpha u(c_1) + \beta u(c_2),$$

where α and β are weights on goods 1 and 2 relative to good 0.

Consumer Optimization: Algebraic Analysis

The consumer chooses c_0, c_1, and c_2 to maximize the utility function

$$u(c_0) + \alpha u(c_1) + \beta u(c_2),$$

subject to the budget constraint

$$Y \geq p_0 c_0 + p_1 c_1 + p_2 c_2.$$

The Lagrangian for this problem is

$$L = u(c_0) + \alpha u(c_1) + \beta u(c_2) + \lambda(Y - p_0 c_0 - p_1 c_1 - p_2 c_2).$$

Consumer Optimization: Algebraic Analysis

$$L = u(c_0) + \alpha u(c_1) + \beta u(c_2) + \lambda(Y - p_0 c_0 - p_1 c_1 - p_2 c_2).$$

First-order conditions:

$$u'(c_0^*) - \lambda^* p_0 = 0$$

$$\alpha u'(c_1^*) - \lambda^* p_1 = 0$$

$$\beta u'(c_2^*) - \lambda^* p_2 = 0$$

Consumer Optimization: Algebraic Analysis

The first-order conditions

$$u'(c_0^*) - \lambda^* p_0 = 0$$
$$\alpha u'(c_1^*) - \lambda^* p_1 = 0$$
$$\beta u'(c_2^*) - \lambda^* p_2 = 0$$

imply

$$\frac{u'(c_0^*)}{\alpha u'(c_1^*)} = \frac{p_0}{p_1} \text{ and } \frac{u'(c_0^*)}{\beta u'(c_2^*)} = \frac{p_0}{p_2} \text{ and } \frac{\alpha u'(c_1^*)}{\beta u'(c_2^*)} = \frac{p_1}{p_2}.$$

The marginal rate of substitution equals the relative prices.

Consumer Optimization: The Time Dimension

Irving Fisher (US, 1867-1947) was the first to recognize that the basic theory of consumer decision-making could be used to understand how to optimally allocate spending intertemporally, that is, over time, as well as how to optimally allocate spending across different goods in a static, or point-in-time, analysis.

Consumer Optimization: The Time Dimension

Following Fisher, return to the case of two goods, but reinterpret:

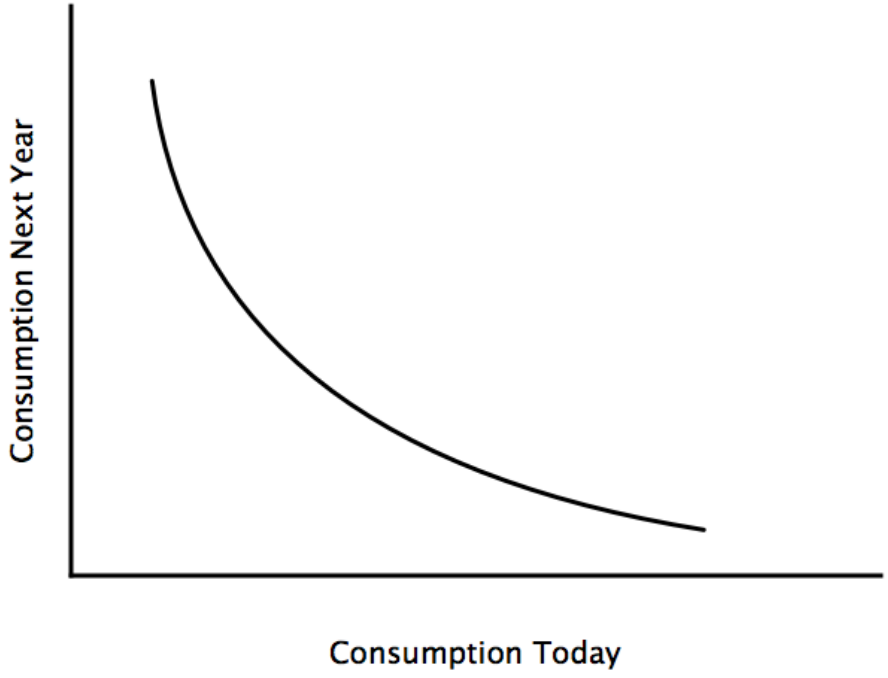
$$c_0 = \text{consumption today}$$
$$c_1 = \text{consumption next year}$$

Suppose that the consumer's utility function is

$$u(c_0) + \beta u(c_1),$$

where β now has a more specific interpretation, as the discount factor, a measure of patience.

Consumer Optimization: The Time Dimension



A concave utility function implies that indifference curves are convex, so that the consumer has a preference for a smoothness in consumption.

Consumer Optimization: The Time Dimension

Next, let

$$Y_0 = \text{income today}$$
$$Y_1 = \text{income next year}$$
$$s = \text{amount saved (or borrowed if negative) today}$$
$$r = \text{interest rate}$$

Consumer Optimization: The Time Dimension

Today, the consumer divides his or her income up into an amount to be consumed and an amount to be saved:

$$Y_0 \geq c_0 + s.$$

Next year, the consumer simply spends his or her income, including interest earnings if s is positive or net of interest expenses if s is negative:

$$Y_1 + (1 + r)s \geq c_1.$$

Consumer Optimization: The Time Dimension

Divide both sides of next year's budget constraint by $1 + r$ to get

$$\frac{Y_1}{1+r} + s \geq \frac{c_1}{1+r}.$$

Now combine this inequality with this year's budget constraint

$$Y_0 \geq c_0 + s.$$

to get

$$Y_0 + \frac{Y_1}{1+r} \geq c_0 + \frac{c_1}{1+r}.$$

Consumer Optimization: The Time Dimension

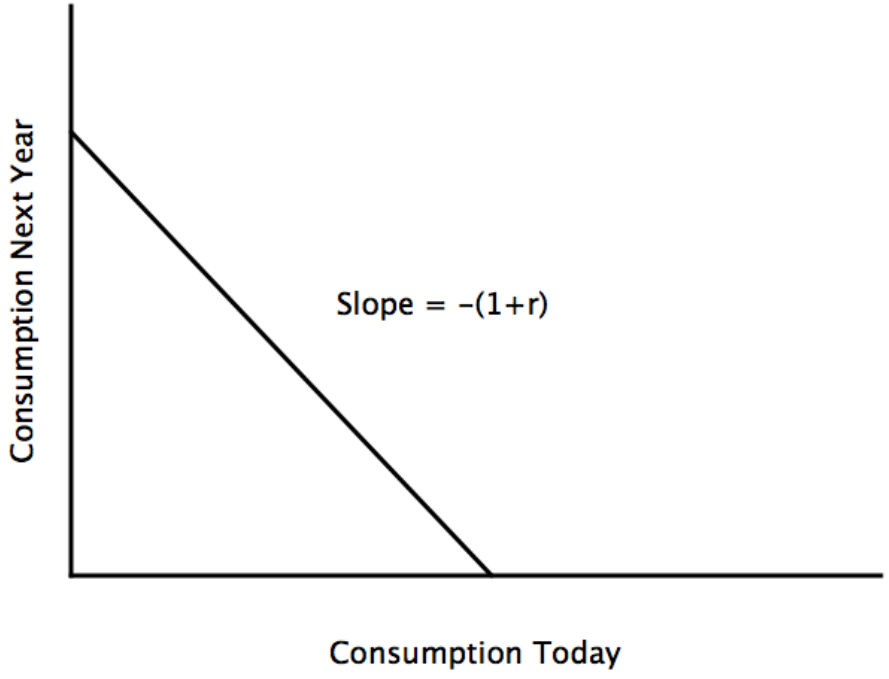
The "lifetime" budget constraint

$$Y_0 + \frac{Y_1}{1+r} \geq c_0 + \frac{c_1}{1+r}$$

says that the present value of income must be sufficient to cover the present value of consumption over the two periods. It also shows that the "price" of consumption today relative to the "price" of consumption next year is related to the interest rate via

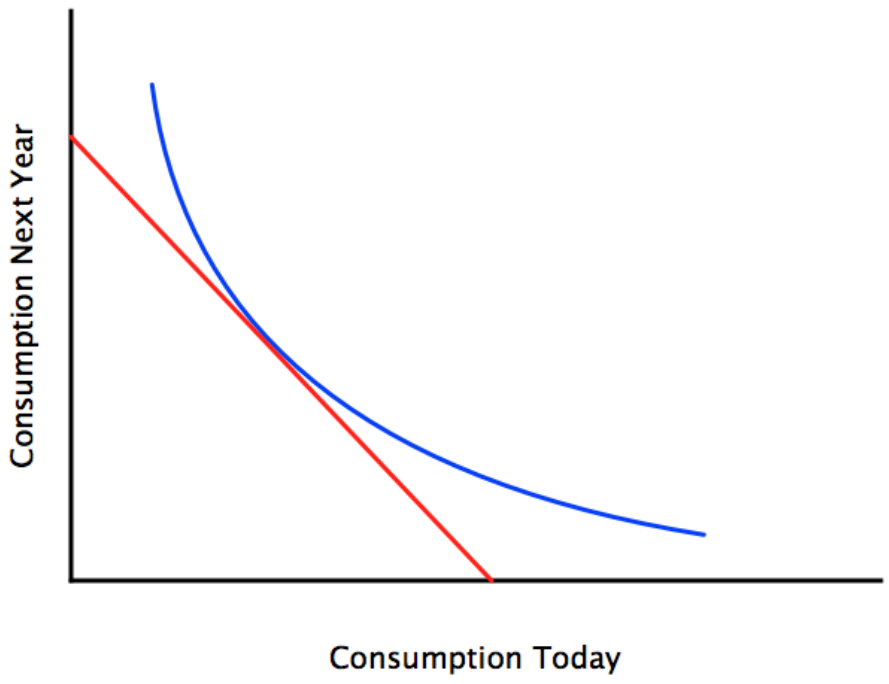
$$\frac{p_0}{p_1} = 1 + r.$$

Consumer Optimization: The Time Dimension



The slope of the intertemporal budget constraint is $-(1+r)$.

Consumer Optimization: The Time Dimension



At the optimum, the intertemporal marginal rate of substitution equals the slope of the intertemporal budget constraint.

Consumer Optimization: The Time Dimension

We now know the answer ahead of time: if we take an algebraic approach to solve the consumer's problem, we will find that the IMRS equals the slope of the intertemporal budget constraint:

$$\frac{u'(c_0)}{\beta u'(c_1)} = 1 + r.$$

But let's use calculus to derive the same result.

Consumer Optimization: The Time Dimension

The problem is to choose c_0 and c_1 to maximize utility

$$u(c_0) + \beta u(c_1)$$

subject to the budget constraint

$$Y_0 + \frac{Y_1}{1+r} \geq c_0 + \frac{c_1}{1+r}.$$

The Lagrangian is

$$L = u(c_0) + \beta u(c_1) + \lambda \left(Y_0 + \frac{Y_1}{1+r} - c_0 - \frac{c_1}{1+r} \right).$$

Consumer Optimization: The Time Dimension

$$L = u(c_0) + \beta u(c_1) + \lambda \left(Y_0 + \frac{Y_1}{1+r} - c_0 - \frac{c_1}{1+r} \right).$$

The first-order conditions

$$u'(c_0^*) - \lambda^* = 0$$

$$\beta u'(c_1^*) - \lambda^* \left(\frac{1}{1+r} \right) = 0.$$

lead directly to the graphical result

$$\frac{u'(c_0^*)}{\beta u'(c_1^*)} = 1 + r.$$

Consumer Optimization: The Time Dimension

At first glance, Fisher's model seems unrealistic, especially in its assumption that the consumer can borrow at the same interest rate r that he or she receives on his or her savings.

A reinterpretation of saving and borrowing in this framework, however, can make it more applicable, at least for some consumers.

Investment Strategies and Cash Flows

Investment Strategy	Cash Flow at $t = 0$	Cash Flow at $t = 1$
Saving	-1	$+(1+r)$
Buying a bond (long position in bonds)	-1	$+(1+r)$

Investment Strategies and Cash Flows

Investment Strategy	Cash Flow at $t = 0$	Cash Flow at $t = 1$
Borrowing	$+1$	$-(1 + r)$
Issuing a bond	$+1$	$-(1 + r)$
Short selling a bond (short position in bonds)	$+1$	$-(1 + r)$
Selling a bond (out of inventory)	$+1$	$-(1 + r)$

Investment Strategies and Cash Flows

Investment Strategy	Cash Flow at $t = 0$	Cash Flow at $t = 1$
Buying a stock (long position in stocks)	$-P_0^s$	$+P_1^s$
Short selling a stock (short position in stocks)	$+P_0^s$	$-P_1^s$
Selling a stock (out of inventory)	$+P_0^s$	$-P_1^s$

Consumer Optimization: The Time Dimension

Someone who already owns bonds can "borrow" by selling a bond out of inventory. In fact, theories like Fisher's work better when applied to consumers who already own stocks and bonds.

Greg Mankiw and Stephen Zeldes, "The Consumption of Stockholders and Nonstockholders," *Journal of Finance*, 1991.

Annette Vissing-Jorgensen, "Limited Asset Market Participation and the Elasticity of Intertemporal Substitution," *Journal of Political Economy*, 2002.